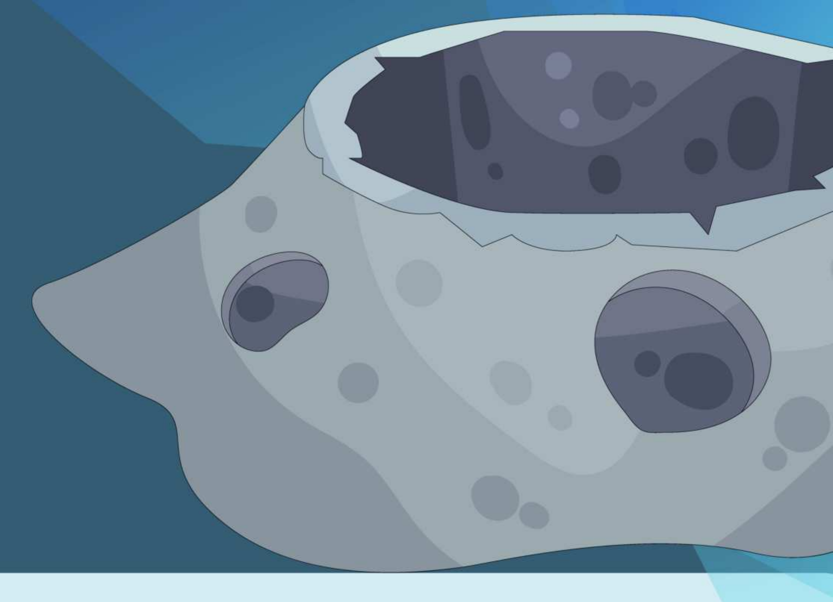
# Advanced Orbit Calculator: Non-Uniform Density of Asteroids and their Satellite Parameters



## Introduction

16 psyche is an asteroid within a large belt of asteroids between Mars and Jupiter. One of the first missions launched by the National Aeronautics Space Administration (NASA) in 2023 found that the density estimate of 16 Psyche is within a numerical range (3,400-4,100 kg/m³).

This range is shown by the uneven craters of the M-type asteroid (has a metallic rich core of pure nickel and iron metal).

Figure 1: Image of 16 Psyche taken by NASA (NASA,2023) .

Asteroids are typically frowned upon to test and work with because of their irregular shape and unevenness, however, it is good to model density because of the drastic changes in mass and porous areas. 16 Psyche itself is 10-20% porous.

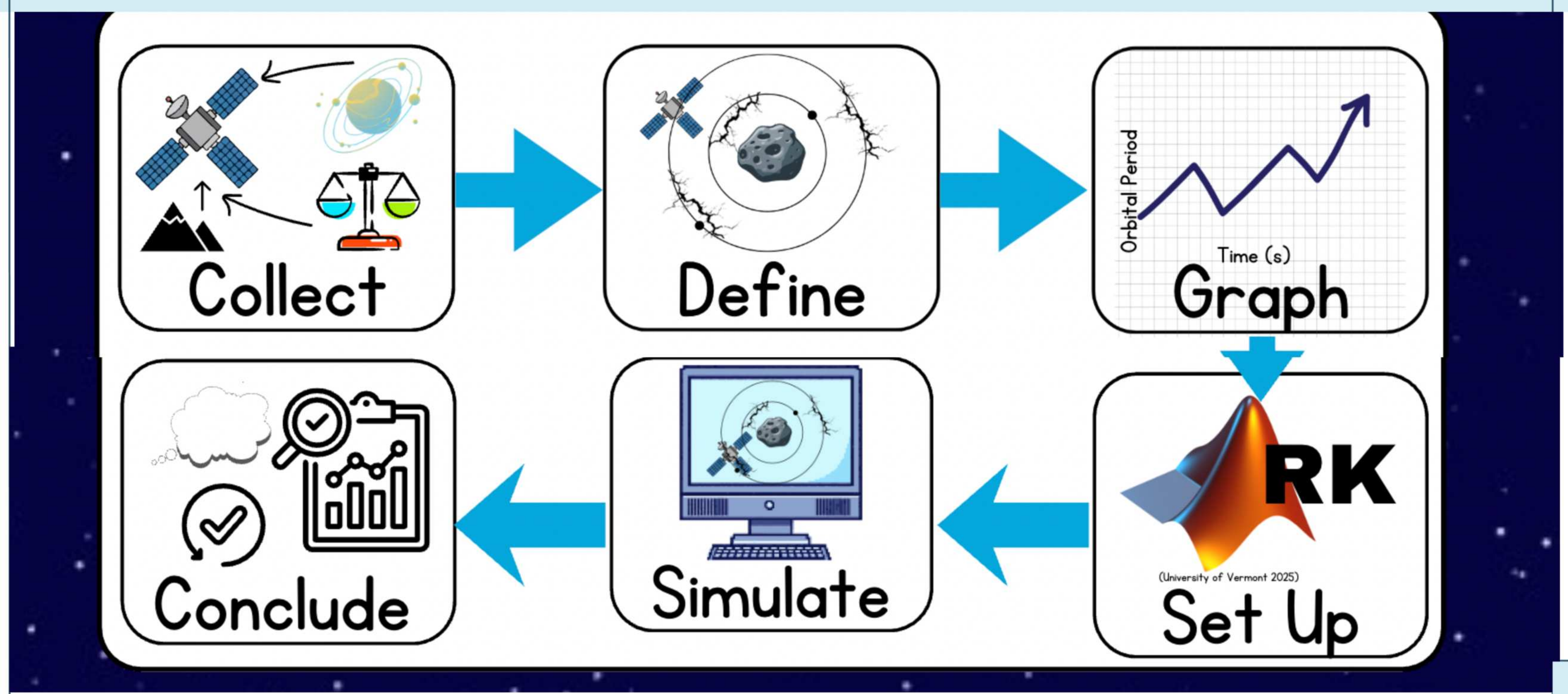
## Purpose

Objective: To create a proof of concept model that finds out the relationship between the nonuniform density of 16 Psyche and satellite parameters (orbital velocity and orbital period).

16 Psyche has possibilities to be an economical hub for scientists in the future due to its core contents. Figuring out the impact of this nonuniform density distribution could help scientists prevent collisions and premature reentries of satellites to Earth. There has not been a previously curated solution for this specific problem.

Hypothesis: In a MATLAB simulated setting, if the density distribution changes as a satellite goes across an increased density patch of a celestial body, then the orbital period will decrease and the orbital velocity will increase because the gravitational attraction between the celestial body and satellite will be greater, leading to the gravity pulling the satellite faster.

## Methodology




1. First, satellite data, altitude points, and 16 Psyche constants were collected.
2. Second, the nonuniform density was defined through elliptical orbit equations (ex. vis viva) and the density range was used to calculate the orbital velocity and orbital period at 200 points within the density range.
3. The calculations made in the previous step were graphed to find the relationship between these variables.
4. The Rung Kutta method was set up within the MATLAB simulation and this method found the orbital velocity and orbital period in different steps to reduce the numerical error on the project (sort of life the Euler's Method in AP Calculus BC).
5. The Rung Kutta Method was simulated into a graph to show the elliptical orbit around 16 Psyche.
6. Conclusions were made based on the three graphs curated by the MATLAB simulation.

Constants: radius of 16 Psyche, gravitational constant, density range of 16 Psyche, Altitudes (709, 303, 190, 75km)
Independent Variable: Density varation of 16 Psyche
Dependent Variable: Satellite Parameters (Orbital Velocity + Orbital Period)

## Results

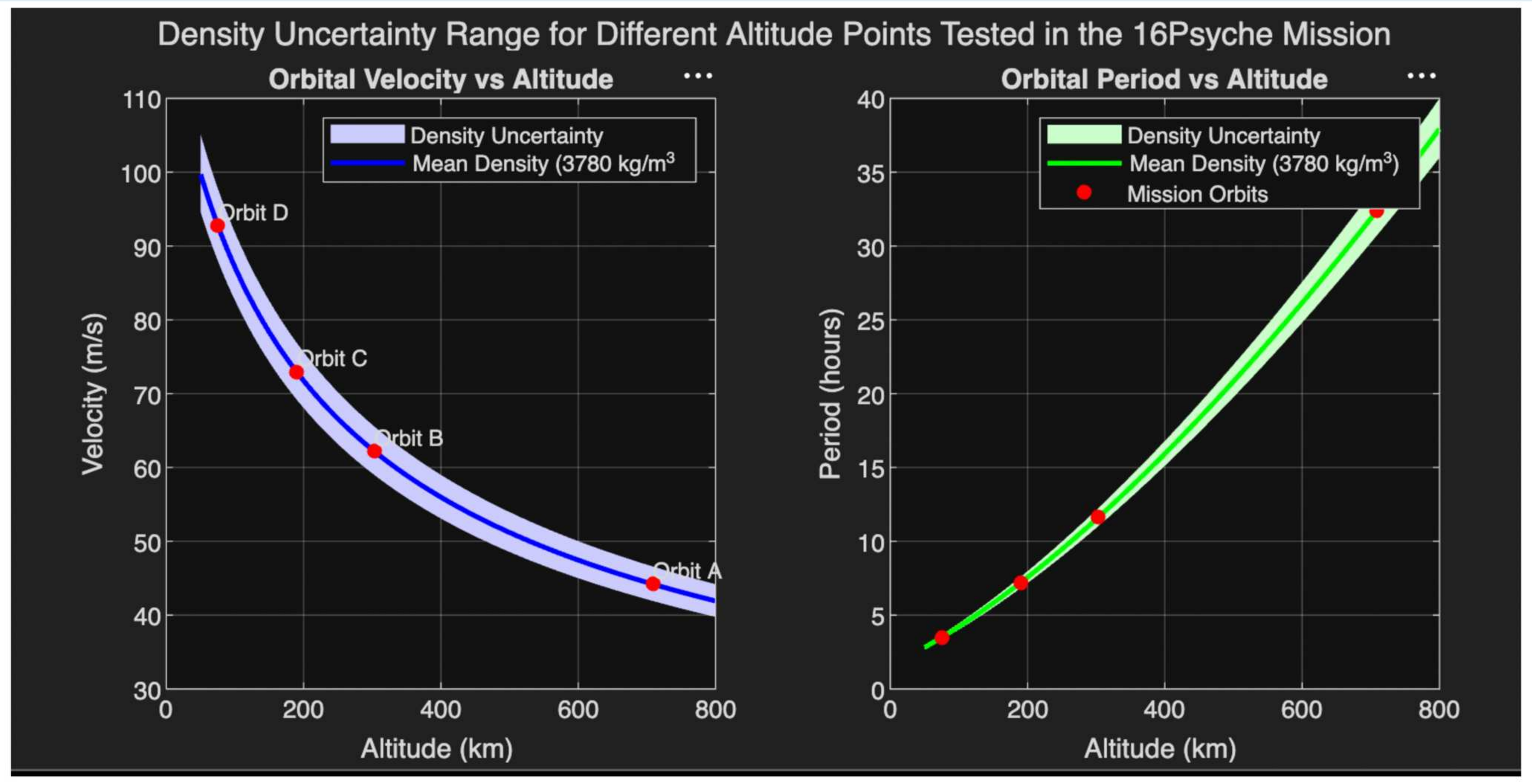



Figure 2: Graphs generated by MATLAB.

## Analysis

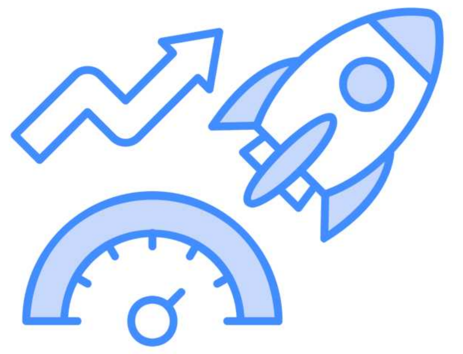
- Density uncertainty band widens at lower altitudes
  - which show the deviations in density distribution at these areas.
- The orbital velocity decreases as the density distribution widens.
- Orbital period graph that the orbital period increases as the altitude increases.
  - The density uncertainty band expands as it goes onto higher altitudes, showing its impact on the orbital period itself.

## Conclusions

This project demonstrated that non-uniform density distributions within asteroid 16 Psyche significantly affect satellite orbital parameters. The results showed that the uncertainty bands widened at lower altitudes, showing that satellites closer to the asteroid are more affected by uneven mass distribution. These findings support the hypothesis that density variations within celestial bodies can meaningfully influence satellite motion.

## Extensions

This project can be extended in the future by adding orbital acceleration. The asteroid itself can be changed based on the asteroid properties to make it a more widely used model.



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